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1. Code/Output:

```
. gen lwage = log(wage)
```

```
. reg lwage educ exper hrswk
```

Source	SS	df	MS	Number of obs	=	4,838
Model	404.898414	3	134.966138	F(3, 4834)	=	548.16
Residual	1190.22017	4,834	.246218488	Prob > F	=	0.0000
				R-squared	=	0.2538
				Adj R-squared	=	0.2534
Total	1595.11859	4,837	.329774361	Root MSE	=	.4962

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.0928357	.0026528	35.00	0.000	.087635	.0980364
exper	.0058515	.000566	10.34	0.000	.0047418	.0069611
hrswk	.010701	.000678	15.78	0.000	.0093718	.0120303
_cons	.9753841	.0475269	20.52	0.000	.8822098	1.068558

Estimated equation:

$$\ln(\text{wage}) = 0.9753 + 0.0928\text{EDUC} + 0.0058\text{EXPER} + 0.0107\text{HRSWK}$$

$\beta_2 = 0.0928$: if years of education increased by 1, then the log of wage would increase

by 0.0928 if experience and hours worked were held constant

$\beta_3 = 0.0058$: if years of experience after education increased by 1, then the log of wage

would increase by 0.0058 if education and hours worked were held constant

$\beta_4 = 0.0107$: if the number of hours worked per week increased by 1, then the log of

wage would increase by 0.0107 if experience and education were held constant

Looking at the $P > |t|$ column (p-values), all of the variables are significant at

$\alpha = 0.05, 0.01, 0.1$, etc. This suggests that all are useful in predicting $\ln(\text{wage})$, and while

they are all close to 0, none of them have 95% confidence intervals which contain 0,

further suggesting that none of the parameters are 0. If the parameters were 0, it would

mean that they are insignificant in predicting the value of $\ln(\text{wage})$

2.

Hypotheses: $H_0: \beta_2 \geq 0.1$ vs. $H_1: \beta_2 < 0.1$

$$\text{Test Statistic: } \frac{\hat{\beta}_2 - \beta_{2(0)}}{se(\hat{\beta}_2)} = \frac{.0928357 - 0.1}{0.0026528} = -2.7006$$

Critical Value / Rejection Region: $-t_{0.05, 4834} = -1.645$

Conclusion: The test statistic is less than the critical value, so it falls within the rejection region. Thus, the null hypothesis is rejected at $\alpha = 0.05$, meaning an extra year of education does not increase wage rate by at least 10 percent.

3. 90% interval estimate for percentage increase in wage from working an additional hour per week.

$$b_4 \pm t_{0.05, 4834}(se(b_4)) = 0.010701 \pm 1.645(0.000678) = (0.00958569, 0.01181631)$$

4.

Code/output:

```
. gen educsq = educ*educ
. gen expersq = exper*exper
. reg lwage educ exper hrswk educsq expersq
```

Source	SS	df	MS	Number of obs	=	4,838
Model	443.124998	5	88.6249996	F(5, 4832)	=	371.73
Residual	1151.99359	4,832	.238409269	Prob > F	=	0.0000
				R-squared	=	0.2778
				Adj R-squared	=	0.2771
Total	1595.11859	4,837	.329774361	Root MSE	=	.48827

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.0664795	.0139851	4.75	0.000	.0390623	.0938966
exper	.0317899	.0021415	14.84	0.000	.0275915	.0359882
hrswk	.0091276	.000679	13.44	0.000	.0077965	.0104587
educsq	.0009018	.0004941	1.83	0.068	-.0000668	.0018703
expersq	-.0004792	.0000381	-12.57	0.000	-.0005539	-.0004045
_cons	.9504657	.1059524	8.97	0.000	.7427507	1.158181

The parameters associated with education, experience, hours worked, and experience squared are all significantly different from 0 at any $\alpha > 0.000$, while education squared is only significantly different from 0 at $\alpha > 0.068$

5.

$$d\ln(WAGE)/dEDUC = d(B_1 + B_2educ + B_3exper + B_4hrswk + B_5educ^2 + B_6exper^2 + e)$$

$$d(wage)(1/wage)/dEDUC = 0 + educ + 0 + 0 + 2B_5educ$$

$$\text{Marginal effect: } (\overline{EDUC} + 2B_5\overline{EDUC}) \overline{WAGE}$$

$$d\ln(WAGE)/dEXPER$$

$$\text{Marginal effect: } (\overline{EXPER} + 2B_6\overline{EXPER}) \overline{WAGE}$$